A Machine Vision-based Realtime Anomaly Detection Method for Industrial Products Using Deep Learning

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***Abstract*—In the process of industrial production, anomaly detection is the key link to ensure the high quality of the product. This paper deeply studies the method of anomaly detection for industrial products based on deep learning. For the balanced image data set of industrial production products, this paper proposes a supervised anomaly detection model based on YOLOv3. This model constructs the ROI classifier to detect anomaly types. For the unbalanced image data set (only few anomaly images) of industrial production products, this paper proposes a semi-supervised anomaly detection model based on Fast-AnoGAN. This model is built from normal samples only. It uses the trained WGAN-GP model to generate images, and achieves anomaly detection by monitoring the anomaly score which is obtained by calculating the difference between the generated image and the test image. The two proposed anomaly detection models evaluated on both balanced and unbalanced data sets in the real industrial production scenarios. The performance evaluation results demonstrate that the two proposed anomaly detection models based on deep learning can well meet the dual requirements of real-time and accuracy for anomaly detection in the high-speed industrial production scenarios.**

Keywords—Industrial production, Machine Vision, Anomaly Detection, Deep Learning, YOLOv3, Fast-AnoGAN

I. INTRODUCTION

With the advent of Industry 4.0, improving the anomaly detection performance of product in industrial production is the key to ensuring product quality and achieving intelligence manufacturing. Nowadays, the detection technology based on machine vision has become more and more mature. With the strengths of high accuracy, low cost and nondestructive, it has been widely studied and applied in the field of industrial detection instead of manual detection.

The traditional machine learning detection method needs feature extraction and classifier. Firstly, the industrial production product images are preprocessed, and then the corresponding features are extracted thereafter to perform classification detection with classifiers[1]. Both stages require the participation of professionals in relevant fields to design feature extractors and classifiers manually[2]. Moreover, in the high-speed and complex industrial production scenario, the traditional machine vision detection technology has the disadvantages of complex model parameters settings, poor robustness and poor adaptability, which leads to the failure to meet the requirement of real-time speed and high precision in industrial production.

In recent years, deep learning has been deeply studied and widely applied in the fields of computer vision due to its powerful feature learning ability. The deep learning-based detection method integrates feature learning into the process of building model. The deep network model automatically learns deeper feature through massive data instead of designing feature extractor. Moreover, the performance of detection models based on deep learning is superior to that based on traditional machine learning. It brings new ideas to product anomaly detection under high-speed and complex industrial production scenarios. However, the balanced data set has to collect a quantity of anomaly instances. This is especially difficult in scenarios where anomaly occur very rarely like industrial production[3]. Moreover, manual labeling of anomaly is extremely time-consuming and laborious[4]. So, there are a large number of unbalanced product image data sets in industrial production scenarios.

This paper explores suitable deep learning approaches applied to anomaly detection of the industrial production product. For the balanced image data set of industrial production product which provides both sufficient normal and anomaly samples during model training, we propose a supervised anomaly detection model based on YOLOv3[5]. This model uses a new basic network Darknet-53 which simplifies the complexity of network training and improves the detection performance compared with other networks like Darknet-19. It better guarantees the model with high precision and real-time performance. For unbalanced data set, we consider the unsupervised learning model GAN[6] (Generative Adversarial Networks) for image generation and propose the Fast-AnoGAN improved from AnoGAN[7] to build a semi-supervised anomaly detection model that is suitable for data sets with a small number of anomaly samples. This model only requires normal samples for training and detects anomaly by monitoring the anomaly score which is obtained through calculating the difference between the generate image and the test image. The performance is verified in a real industrial production scenario regarding both speed and accuracy.

The remainder of this paper is structured as follows. Section II reviews the related algorithms and models on deep

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learning-based anomaly detection. Section III describes the details of the two anomaly detection models we have proposed. Section IV focus on the experiments to evaluate the performance of the two anomaly detection models. Section V concludes this paper.

II. RELATED WORK

Due to the diversity of industrial production product data sets, among the anomaly detection methods based on deep learning can distinguish between two groups: multi-class learning based anomaly detection method and one-class learning based anomaly detection method[8].

A. Multi-class learning based anomaly detection method

The multi-class learning based anomaly detection method uses both normal and anomaly samples during training. This method can direct application of the object detection methods based on deep learning to locate and classify anomalies in images.

The object detection methods based on deep learning can mainly be categorized into two types[9]. The region proposal-based method follows traditional object detection pipeline, generating region proposals at first and then classifying each proposal into different object categories. While the end-to-end learning method regards object detection as a regression or classification problem, adopting a unified framework to achieve final results (classes and locations) directly. The region proposal-based method mainly includes R-CNN[10], Fast R-CNN[11], Faster R-CNN[12], and R-FCN[13]. Cha et al.[14] developed a structural damage detection method based on Faster R-CNN to detect five types of surface damages. The end-to-end learning method mainly includes YOLO[15], SSD[17], YOLOv2[16] and YOLOv3[5]. Li, Yiting, et al.[18] presents a surface defect detection method based on SSD

As shown in the Table I, although the region proposal-based method has higher detection accuracy, the high computational complexity of the region proposals extraction process makes its detection speed difficult to meet the real-time requirements. So, it cannot be well applied to industrial production. However, the end-to-end learning method removes the region proposals extraction step, uses the object detection as a regression problem to directly return the position of the object from the input image and output classification result, which has a huge increase in the detection speed, meeting the real-time requirements.

TABLE I. THE PERFORMANCE COMPARISON OF VARIOUS OBJECT DETECTION MEHTODS[9]

Detection Method	Train	MAP	FPS
R-CNN	VOC 07	66%	0.03
Fast R-CNN	VOC 07	70.0%	0.5
Faster R-CNN(VGG16)	VOC 07	73.2%	9.1
Faster R-CNN(ResNet101)	VOC 07	83.8%	0.4
YOLO	VOC 07	63.4%	45
R-FCN(ResNet101)	VOC 07	83.6%	5.9
SSD300	VOC 07	74.3%	46
YOLOv2(352*352)	VOC 07	73.7%	81

With the development of research in recent years, the end-to-end learning method has also greatly improved the detection accuracy, making it possible to apply the object detection method base on deep learning in industrial production.

B. One-class learning based anomaly detection method

The one-class learning based anomaly detection method only requires normal samples for training. This method requires to train a generator with the normal image completion or generation capabilities. Through comparing the difference between the generate image and the test image to detect anomaly.

The main generator model includes Autoencoder, Deep Belief Network (DBN), Generative Adversarial Network (GAN). GAN[6] as the unsupervised learning structure model has been widely concerned and applied since its introduction in 2014, and has become one of the most promising methods of generation in recent years.

The GAN consists of two parts, shown in Fig.1. A part of the network model is a discriminator (D), which is used to distinguish whether the input data is real data or generated fake sample. Another part of the network model is a generator (G), which is used to generate random fake samples that approximate real data. GAN introduces the confrontation mechanism to learn the data distribution of the training set through the confrontation learning of generator (G) and discriminator (D), thus obtaining the generator that learned the corresponding generation mechanism. GAN is well suited for modeling high-dimensional complex distributions of real data. The advantage is that it can use a small amount of labeled data and a large amount of unlabeled data to train, and get the generator that can generate data similar to real target data[19].

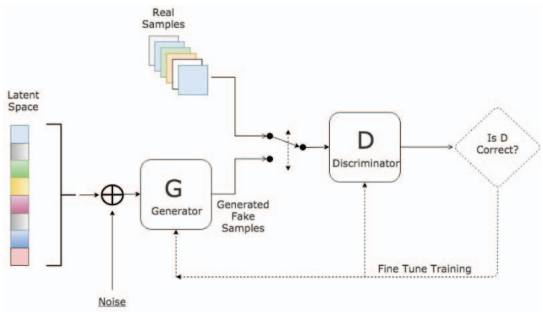

Fig. 1. Generative Adversarial Network

GAN[6] as a pioneering model, has been derived from LSGAN, DCGAN[24], WGAN, WGAN-GP, BEGAN and other improved GAN-based model with the deepening of research, which increases the training speed, improves the stability of the model and promotes the quality of generator generation.

III. ANOMALY DETECTION MODEL

A. Anomaly detection for balanced data

For the balanced data set of industrial production product, we compare different end-to-end learning methods in object detection, and finally build the supervised anomaly detection model based on YOLOv3[5] and its light model Tiny-YOLOv3.

1) Basic Model: YOLOv3

YOLOv3[5] was proposed in 2018 and is based on the improvement of YOLOv2[16]. Unlike the basic network Darknet-19 used by YOLOv2, YOLOv3 uses a new network Darknet-53 for feature extraction. As shown in the Table II, it can be seen that Darknet-53 is superior to Darknet-19,

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ResNet-101 and ResNet-152 in detection performance. Although the speed of detection is not as fast as Darknet-19, it still meets the requirements of real-time.

TABLE II. THE TEST RESULT OF IMAGENET DATA[5]

Backbone	Top-1/%	Top-5/%	FPS
Darknet-19	74.1	91.8	171
ResNet-101	77.1	93.7	53
ResNet-152	77.6	93.8	37
Darknet-53	77.2	93.8	78

In addition, YOLOv3 also introduces the idea of anchor boxes in Faster-RCNN, using three scale predictions for the training datasets, and predicting 3 boxes for each scale. It greatly improves the accuracy of target detection.

In category prediction, YOLOv3 no longer uses the SoftMax used by YOLOv2 to classify each box, but multiple logistic classifiers are used to perform class prediction using binary cross entropy loss during training.

Based on the above, the target detection accuracy of YOLOv3 is further improved compared to other end-to-end learning object detection models such as YOLO, YOLOv2 and SSD. As can be seen from the performance comparison shown in the Table III, while maintaining high accuracy, it has maintained a high detection speed and achieved a perfect balance between detection accuracy and speed.

TABLE III. THE METHODS PERFORMANCE COMPARISON[20]

Detection Method	Input	MAP	FPS
YOLO	448*448	62.52%	42.34
YOLOv2	416*416	73.82%	64.65
YOLOv2	544*544	75.96%	39.14
SSD300	300*300	74.18%	58.78
SSD512	512*512	76.83%	27.75
YOLOv3	416*416	88.09%	51.26

[a] Performance testing using the UA-DETRAC data set

Tiny-YOLOv3 is the lightweight version of YOLOv3. In the feature extraction part, the feature extraction model Darknet-53 used in YOLOv3 is composed of 53 convolution layers and 5 maximum pooling layers, while Tiny-YOLOv3 simplifies the feature extraction model into 15 layers, which are composed of 10 convolution layers and 5 maximum pooling layers. And in the scale prediction part, YOLOv3 finally has 3 branch outputs for prediction, and each feature map uses 3 anchors for prediction. While in Tiny-YOLOv3, there are finally 2 branches for prediction, and each feature map uses 3 anchors for prediction[21].

Based on the above, Tiny-YOLOv3 simplifies the feature extraction and scale prediction process based on YOLOv3, and greatly improves the recognition speed while maintaining high recognition accuracy. It's more suitable for high-speed, real-time industrial production.

2) Model detect flowchart

The flowchart of the YOLOv3 anomaly detection model is shown in Fig.2. We initialize the YOLOv3 network model and train the model by preprocessing the training set with resize and grayscale as input. Final, we will get a trained YOLOv3 model after enough training.

During the test, we do the same preprocessing for test set and input the trained YOLOv3 model. The model will output the class and location of the detected object. If any object is labeled as anomaly, the detected image is discriminated as an abnormal image, otherwise it is the normal image.

The flowchart of the Tiny-YOLOv3 anomaly detection model is the same, except that the YOLOv3 network model is changed to Tiny-YOLOv3 network model.

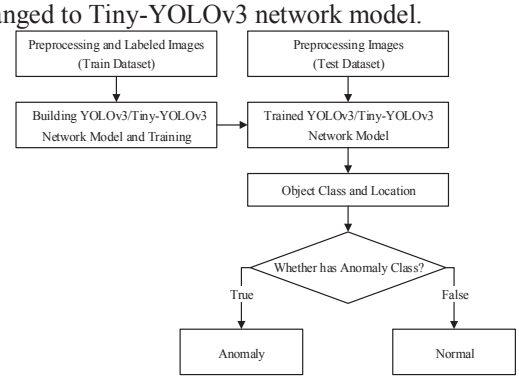

Fig. 2. The flowchart of the YOLOv3 anomaly dectection model

B. Anomaly detection for unbalanced data

1) Basic Model: WGAN-GP[23]

For the unbalanced data set of industrial production product, we build the semi-supervised anomaly detection model based on Fast-AnoGAN, which improves on the basis of AnoGAN proposed by Schlegl et al[7].

We use WGAN-GP model to train the generator and the discriminator instead of DCGAN[24] model used in AnoGAN. WGAN-GP is an improved version of WGAN[22] (Wasserstein GAN). WGAN-GP uses Wasserstein distance to replace the JS divergence in GAN, which solves the problem of model collapse, improves the stability of training. What's more, WGAN-GP uses gradient punishment to satisfy the Lipschitz continuity condition, and solves the problem of training gradient disappearance and gradient explosion in WGAN. So, compared with DCGAN, WGAN-GP has faster training speed and can generate higher quality images[25].

The structure of the generator and the discriminator network structures we use in the model are shown in Fig.3 and Fig.4. During the training, we input the n-dimensional latent noise z_n and the preprocessed normal image. The generator can learn the distribution of the normal dataset. The trained generator will generate an image $G(z_n)$ that is similar to the normal image when we input the n dimensional latent noise z_n.

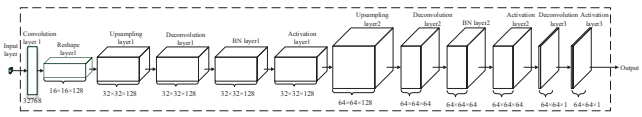

Fig. 3. Generator Structure

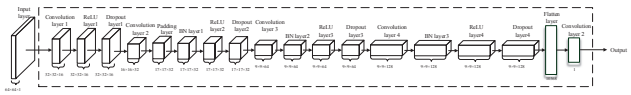

Fig. 4. Discriminator Structure

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2) Model detect flowchart

The Fast-AnoGAN anomaly detection model flowchart is shown in Fig.5. This model determines whether the detected image is abnormal by calculating anomaly score between the generated image and the detected image. It eliminates the time-consuming iterative optimization process in the AnoGAN[7], and directly define the anomaly score calculation formula to calculate the anomaly score as the parameter of abnormality discrimination. When the anomaly score is higher than the normal threshold, the detected image is discriminated as an abnormal image. The normal threshold is the mean value of abnormal scores calculated from a number of known normal data.

The anomaly score calculation formula consists of two parts, $\mathcal{L}_R$ loss and $\mathcal{L}_D$ loss. Where x is the preprocessed detected image after resize and grayscale, z_n is the n-dimensional latent noise.

$$\mathcal{L}(x, z_n) = (1-\lambda)\cdot\mathcal{L}_R(x, z_n) + \lambda\cdot\mathcal{L}_D(x, z_n) \quad (1)$$

The $\mathcal{L}_R$ loss calculates the visual dissimilarity between the detected image x and the generated image $G(z_n)$ through using subtract function.

$$\mathcal{L}_R(x, z_n) = \sum|x - G(z_n)| \quad (2)$$

The $\mathcal{L}_D$ loss defines $F(\cdot)$ function which is the output of an intermediate layer in discriminator structure. Through using subtract function to calculate the discriminator output $F(x)$ and $F(G(z_n)$ by feeding the detected image x and generated image $G(z_n)$ into the discriminator.

$$\mathcal{L}_D(x, z_n) = \sum|F(x) - F(G(z_n))| \quad (3)$$

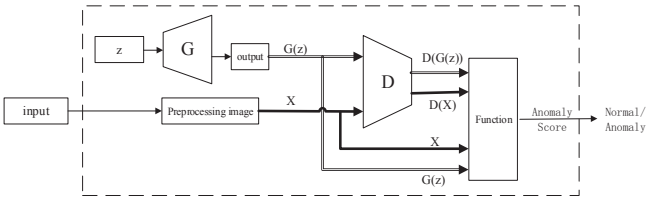

Fig. 5. Fast-AnoGAN Anomaly Detection Model

This model greatly improves the detection rate while ensuring accuracy, thereby realizing abnormal detection of industrially produced products.

IV. EXPERIMENTS

The experimental environment is as follows: deep learning open framework TensorFlow, Windows 10, 8GB RAM, 2.60GHz CPU, GTX 960 GPU and CUDA10.0.132.

A. YOLOv3 anomaly detection model

1) Dataset

We train YOLOv3 and Tiny-YOLOv3 anomaly detection model by using a balanced data set of cigarette transparent tear strip images in the industrial production scenarios for detecting abnormal conditions like deficiency, misalignment and overlength. Fig.6 shows several situations of cigarette transparent tear strip images. The red box area is anomaly and the green box area is normal.

This dataset has 440 images and 880 labeled areas. There are 507 labeled areas are normal and 373 labeled areas are anomaly. So, we have 67 normal images (both labeled areas in the image are normal) and 363 negative sample images (at least one of the labeled areas in images is anomaly). After randomly splitting the dataset, 400 images are used as training set and 40 images as test set.

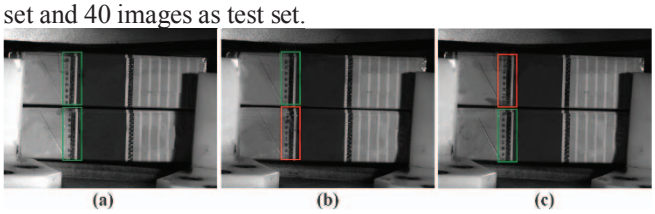

Fig. 6. Several situations of cigarette transparent tear strip images. (a) normal (b)abnomaly: misalignment (c) abnomaly: overlength

2) Testing results

The Fig.7 and Fig.8 respectively show sample test results of the YOLOv3 and Tiny-YOLOv3 anomaly detection model. The first line shows the test images and the second line shows the detection results.

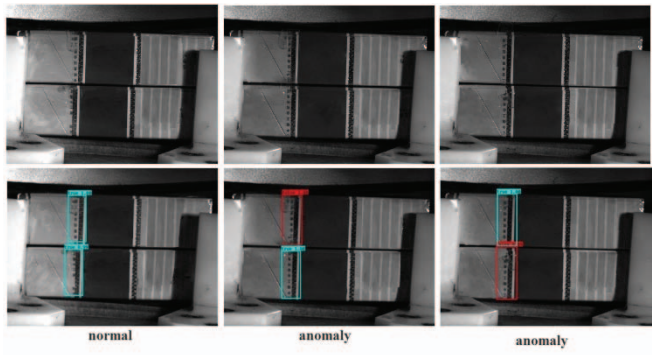

Fig. 7. Sample test results of the YOLOv3 anomaly detection model

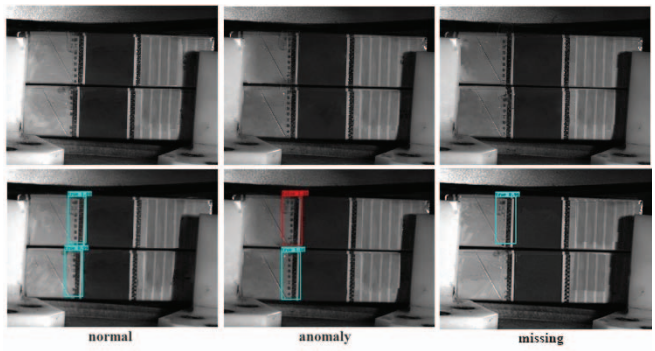

Fig. 8. Sample test results of the Tiny-YOLOv3 anomaly detection model

In order to measure the detection accuracy, we define the detection rate as the success rate of anomaly samples being correctly detected as anomaly and define the missing rate as the rate that the object undetected or class is misjudged. The YOLOv3 and the Tiny-YOLOv3 anomaly detection model performance as Table IV shown. Both YOLOv3 and Tiny-YOLOv3 models have achieved real-time detection speed and high accuracy. The detection speed of Tiny-YOLOv3 is nearly 4 times that of YOLOv3, but its detection accuracy is lower than that of YOLOv3. As shown in Fig.8, the phenomenon of missing detection may happen. So, in the different industrial production scenarios, the Tiny-YOLOv3 anomaly detection model can be used if higher production speed requirements while the YOLOv3 anomaly detection model can be used if high detection accuracy is required.

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TABLE IV. THE MODEL PERFORMANCE

Model	Detection rate	Missing rate	FPS
YOLOv3	100.00%	0.00%	5.74
Tiny-YOLOv3	100.00%	2.50%	22.25

B. *Fast-AnoGAN anomaly detection model*

1) *Dataset*

We train Fast-AnoGAN anomaly detection model by using an unbalanced data set of cigarette box images in the industrial production scenarios for detecting the abnormal condition of the boxed lack, as shown in Fig.9. In order to prevent the interference of background image to model training, the existing registration points in the image are used to crop the detection area as actual training set.

This dataset has 2000 normal images and 4 anomaly images. The test set consists of 36 normal images and 4 anomaly images, and the training set consists of the remaining 1964 normal images.

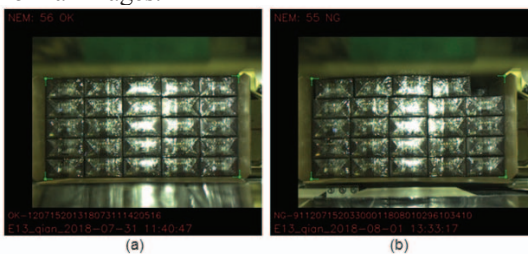

Fig. 9. Cigartte Box Situations
(a) normal (b) abnomaly: boxed lack

2) *Testing results*

For each test image, we use the trained Fast-AnoGAN model to calculate the anomaly score through anomaly score formula, and compared the obtained anomaly score with the normal threshold value to realize the discrimination. As shown in the Table V.

TABLE V. THE RESULT OF SAMPLE TEST IAMGES

N	1	2	3	4
Anomaly score	633.7	643.3	493.3	688.4
Label	anomaly	anomaly	normal	anomaly
N	5	6	7	8
Anomaly score	504.2	699.1	485.1	500.3
Label	normal	anomaly	normal	normal

Detection results is shown in Fig.10. According to the anomaly score formula, we define the x-coordinate is $\mathcal{L}_R$ loss and the y-coordinate is $\mathcal{L}_D$ loss. Each detection image's anomaly score can be represented in this two-dimensional plot. It can be seen that four negative sample images are clearly distinguished as outliers (red points), four from the normal ones.

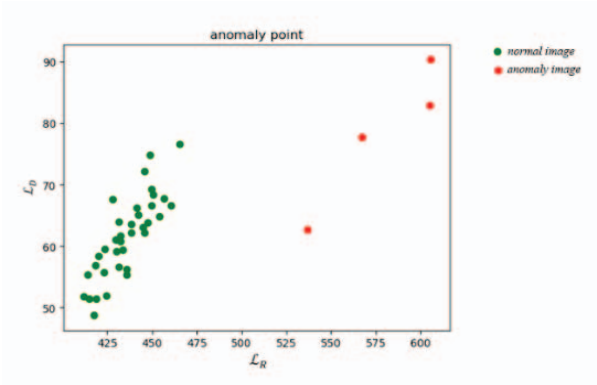

Fig. 10. Anomaly score for the test data set. The green point represents anomaly score of normal image and the red point represent anomaly score of anomaly image

Fig.11 shows the samples of the detection results. The first line is the real test image, the second line is the generated image, and the third line is the defect image which represents the visual result of the difference between the test image and the generated image. The defect image uses the Colormap Jet color mapping method to represent the gray scale of the difference value as the color thermal diagram. In the third line, the missing parts of the anomaly image are shown in blue due to the low gray value.

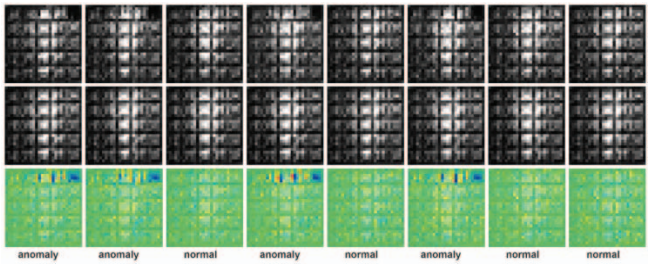

Fig. 11. Sample of the image, reconstructed image and defect image

The anomaly detection performance is shown in Table VI for AnoGAN[7] model and our Fast-AnoGAN model. The definition of detection rate is the same as above. Both two models can accurately detect anomaly, while the detection speed of Fast-AnoGAN is 10 times that of AnoGAN, it has achieved real-time detection speed in the industrial scenarios.

TABLE VI. THE MODEL PERFORMANCE

Model	Detection rate	Speed	FPS
Fast-AnoGAN	100.00%	0.10s	10
AnoGAN	100.00%	1.03s	0.97

V. CONCLUSION

This paper explored anomaly detection methods based on deep learning. For the balanced, we propose the YOLOv3 anomaly detection model. And for unbalanced image dataset, we propose the Fast-AnoGAN anomaly detection model. Both anomaly detection models have high detection accuracy and real-time detection speed in the experiment of cigarette production product images. So, it can be well applied to product anomaly detection in the industrial field.

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In the industrial scene, the variety of production products is diverse, and the production environment is complex and variable. Therefore, transfer learning of the anomaly detection model is required. In further research work, we will consider using the transfer learning combined with the anomaly detection model to speed up the network training speed and improve the generalization performance of the anomaly detection model.

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